

Phase 5- Project Development Phase

Project Title: Script-Controlled ACL – Restrict Record Access Based on Field Value

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The Project Development Phase is the stage in which the design and requirements of the Script-Controlled ACL – Restrict Record Access Based on Field Value project are converted into a functional system. The main purpose of this phase is to implement the access-control mechanism and ensure that records are accessible only when the defined conditions are satisfied.

5.1 System Configuration

The required system environment is prepared before development begins. Tables, records, fields, users, and required access-control components are configured according to the project requirements.

5.2 Record Creation

Different records are created in the system. Each record contains multiple fields with different values. These records provide the data on which the ACL rules will operate.

5.3 Selection of Control Field

A particular field is selected as the control field. The value stored in this field is used by the ACL script to determine whether the requested record should be accessible to the current user.

5.4 User Management

Users are configured with appropriate roles and permissions. When a user requests a record, the system identifies the current user and uses this information during access validation.

5.5 Access Rule Definition

Access rules are defined based on the selected field value. For example, records may be accessible only when the field value matches a predefined condition.

5.6 Script Development

The ACL script is developed to perform the access-control logic. The script retrieves the required field value, evaluates the defined condition, and returns an access decision.

5.7 Record-Level Access Control

The ACL is applied at the record level. This means that access can be controlled separately for individual records instead of giving the same access to every record.

5.8 Access Request Processing

When a user attempts to access a protected record, the request passes through the ACL mechanism. The script checks the user's information and the relevant field value before allowing access.

5.9 Access Granted

When the required condition is satisfied, the ACL allows the user to access the requested record. The user can then view or perform the permitted operation on that record.

5.10 Access Denied

When the condition is not satisfied, the ACL prevents the user from accessing the restricted record. This helps protect sensitive or unauthorized information.

5.11 Error Handling

The system handles invalid, missing, or unexpected field values appropriately. This helps prevent incorrect access decisions and improves the reliability of the access-control mechanism.

5.12 Testing and Debugging

The developed ACL script is tested with different users, records, and field values. Any errors in the script or access conditions are identified and corrected during this stage.

5.13 Security Verification

The system is checked to ensure that unauthorized users cannot bypass the ACL rules. Different access scenarios are tested to verify that the defined restrictions are properly enforced.

5.14 Final Implementation

After successful development and testing, the ACL script is applied to the required records and finalized. The completed system is then ready for demonstration and documentation.

Development Process

System Configuration



Record Creation



Control Field Selection



User Configuration



Access Rule Definition



ACL Script Development



Record-Level Access Control



Testing & Debugging



Final Implementation

Expected Result

The completed system provides a secure, flexible, and automated method of controlling record access. By using field values as access conditions, the system ensures that users can access only the records permitted by the defined ACL rules.

If you want, I can:

Explain key script logic used in the ACL for controlling record access

Describe testing scenarios used to verify ACL script functionality

Detail error handling mechanisms for invalid or missing field values in ACL